

IL21R Deficiency (Immunodeficiency 56)

— Comprehensive Disease Characterization

Autonomous literature-based discovery report. Evidence types are labeled: [human clinical], [model organism], [in vitro], [computational/database]. Primary citations are given as PMIDs.

Summary (Answer to the Research Question)

IL21R Deficiency is an **ultra-rare, autosomal-recessive combined immunodeficiency (Immunodeficiency 56; OMIM #615207)** caused by **biallelic loss-of-function mutations in *IL21R*** (chromosome 16p12.1). Loss of a functional IL-21 receptor abolishes IL-21 → common-gamma-chain/JAK1–JAK3 → STAT3 (and STAT1/STAT5) signaling, crippling B-cell class switching/plasma-cell differentiation, T-follicular-helper help, CD8 memory, and NK/MAIT/NKT cytotoxicity. Clinically this produces recurrent bacterial/fungal/viral infections, hypogammaglobulinemia with paradoxically elevated IgE/atopy, and—most characteristically—**chronic *Cryptosporidium* infection driving sclerosing cholangitis and progressive liver disease**. Hematopoietic stem cell transplantation (HSCT) is the only curative therapy but must be performed before irreversible biliary/liver damage; overall prognosis without early transplant is poor.

First	described:	Kotlarz	et	al.,	<i>J</i>	<i>Exp</i>	<i>Med</i>	2013	(		)
										P 23440042	
Largest	cohort:	Cagdas	et	al.				2021,	n=13	(	)
										P 33929673	

1. Disease Information

Overview. A monogenic inborn error of immunity in the "combined immunodeficiency (CID)" category, defined by absent/non-functional IL-21 receptor signaling. It affects both adaptive (B and T cell) and innate (NK) immunity and characteristically predisposes to opportunistic *Cryptosporidium* infection of the biliary tree. **[human clinical]**

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Key identifiers (computational/database): - **OMIM:** #615207 — "Immunodeficiency 56" (IMD56) - **Gene:** *IL21R*, OMIM 605383; *HGNC*:6006; *NCBI Gene* 50615; *Ensembl* ENSG00000159374; *UniProt* Q9HBE5 (*IL21R_HUMAN*); *cytoband* 16p12.1 - *Orphanet*: Listed under combined immunodeficiencies / *IL-21R-related*; no widely used dedicated *ORPHAcode*—commonly grouped with "Combined immunodeficiency." - *MONDO*: Suggested *MONDO*:0014219 ("immunodeficiency 56") — to be verified against current *MONDO* release. - *ICD-10*: D81.8 / D81.9 (other/unspecified combined immunodeficiencies). *ICD-11*: 4A01.3Y (combined immunodeficiencies, other). - *MeSH*:* No dedicated descriptor; indexed under "Severe Combined Immunodeficiency"/"Primary Immunodeficiency Diseases" + "Receptors, Interleukin-21."

Synonyms / alternative names: IL-21R deficiency; Interleukin-21 receptor deficiency; Immunodeficiency 56 (IMD56); IL21R-related combined immunodeficiency.

Information source. Aggregated disease-level knowledge derived from **individual patient case series** (≈20+ patients worldwide) synthesized in cohort papers and reviews—not from large registries/EHR. **[human clinical]**

2. Etiology

Disease causal factors. *Monogenic/genetic.* Biallelic (homozygous or compound-heterozygous) loss-of-function variants in *IL21R* are necessary and sufficient. An **infectious trigger** (chronic *Cryptosporidium parvum/hominis* biliary infection) is the principal driver of the signature organ pathology (cholangitis/liver disease) in the setting of the genetic immune defect—an

obligate gene × pathogen interaction. [human clinical]
 (P 23440042) :

"cryptosporidial infections associated with chronic cholangitis and liver disease")

Genetic risk factors. - *Causal variants:* e.g., c.G602T (p.Arg201Leu, missense causing receptor mistrafficking/loss of ligand binding); c.240_245delCTGCCA (p.C81_H82del, in-frame deletion). 8 unique mutations reported across 8 families by 2021 (missense, in-frame indels, and other LOF classes). [human clinical/in vitro]

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- *Modifier genes:* None established (cohort too small).

Environmental risk factors. **Consanguinity** (parental relatedness) is the dominant epidemiologic risk factor, increasing homozygosity for rare recessive alleles. Exposure to *Cryptosporidium* (contaminated water/food) converts the immune defect into life-threatening cholangiopathy. No sex, occupational, or toxin risk factors. [human clinical]

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Protective factors. No genetic protective/modifier alleles identified. Environmentally, **avoidance of *Cryptosporidium* exposure** (water precautions/filtration) and early curative HSCT are protective against the worst outcomes. [human clinical] (inferred)

Gene–environment interactions. The core GxE interaction: *IL21R*-null immune state + *Cryptosporidium* exposure → sclerosing cholangitis. Restoration of immune competence (HSCT) enables clearance of the parasite, halting the environmental driver. [human clinical]

(P 30850087) :

cholangiopathy improvement *"following establishment of immune competence"*)

3. Phenotypes (with HPO suggestions & frequencies)

Frequencies from the 13-patient cohort
 (P 33929673)

unless noted. [human clinical]

Phenotype (type)	Frequency	HPO
Recurrent bacterial infections (clinical sign)	84.6%	HP:0002718
Recurrent fungal infections	46.2%	HP:0009098
Recurrent viral infections (incl. CMV)	38.5%	HP:0004429
Cryptosporidiosis (protozoan infection)	~46%	HP:0002726 (recurrent protozoan infection)
Sclerosing cholangitis / cholangitis (sign)	46.2%	HP:0100574
Chronic diarrhea (symptom)	frequent	HP:0002028 / HP:0002014
Chronic liver disease / cirrhosis (sign)	frequent, progressive	HP:0001394
Failure to thrive (sign)	common	HP:0001508
Asthma (sign)	23.1%	HP:0002099
Inflammatory/eczematous skin disease	15.3%	HP:0000964 / HP:0011123
Recurrent anaphylaxis	7.9%	HP:0100845
Lymphadenopathy	reported	HP:0002716
Marginal zone B-cell lymphoma (malignancy)	reported ( 33966600)	HP:0012190
Hypogammaglobulinemia (lab)	most	HP:0004313
Elevated serum IgE (lab)	~50%	HP:0003212
Decreased memory B cells (lab)	most	HP:0002850
Reduced circulating Tfh (lab)	most	(Abnormal Tfh; HP:0410276-family)
Reduced MAIT cells (lab)	most	—
Abnormal/reduced terminally differentiated NK cells (lab)	most	HP:0040218

Characteristics. Onset pediatric — **median 2.5 years (range 0.5–7)**. Severity variable but the hepatobiliary component is typically **progressive** and life-limiting. Infections are **recurrent/chronic**; atopy (IgE, asthma, anaphylaxis) is **episodic**.

Quality-of-life impact. No formal EQ-5D/SF-36/PROMIS data. Qualitatively severe: chronic diarrhea/malabsorption → growth failure; progressive cholangiopathy → cholestasis, portal hypertension, end-stage liver disease; recurrent infections and hospitalizations; transplant-related morbidity. **[human clinical]**

Key (quote):

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"The main clinical manifestations were recurrent bacterial (84.6%), fungal (46.2%), and viral (38.5%) infections; cryptosporidiosis-associated cholangitis (46.2%); and asthma (23.1%). Inflammatory skin diseases (15.3%) and recurrent anaphylaxis (7.9%) constitute novel phenotypes."

4. Genetic / Molecular Information

- **Causal gene:** *IL21R* (HGNC:6006; 16p12.1; NCBI 50615; UniProt Q9HBE5). Type I cytokine receptor; heterodimerizes with the common gamma chain (γ_c / IL2RG / CD132).

[computational/database]

- **Pathogenic variants (ACMG):** classified pathogenic/likely pathogenic. Reported types: **missense** (p.Arg201Leu), **in-frame deletion** (p.C81_H82del), and additional LOF alleles (8 unique across 8 families). Loss-of-function is the uniform functional consequence—no gain-of-function or dominant-negative disease described; heterozygous carriers are healthy (recessive). **[human clinical/in vitro]**

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- **Allele frequency:** Biallelic *IL21R* LOF is essentially absent in gnomAD; individual pathogenic alleles are ultra-rare/private. **[computational/database]**
- **Somatic vs germline:** **Germline.** No somatic/oncogenic *IL21R* role in this disease.
- **Functional consequences:** p.Arg201Leu → **aberrant receptor trafficking to the plasma membrane, loss of IL-21 ligand binding, and abrogated STAT1/STAT3/STAT5**

phosphorylation

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"aberrant trafficking of the IL-21R to the plasma membrane, abrogates IL-21 ligand binding, and leads to defective phosphorylation of ... STAT1, STAT3, and STAT5"). **[in vitro]**

- **Modifier genes / epigenetics / chromosomal abnormalities:** None reported; disease is a point-mutation/small-indel monogenic disorder, not a structural/aneuploidy syndrome.

5. Environmental Information

- **Environmental factors:** No toxin/radiation/pollution etiology. The critical environmental element is **exposure to the waterborne protozoan *Cryptosporidium***.
- **Lifestyle factors:** Not applicable as risk drivers; hygiene/water safety are relevant for preventing opportunistic infection.
- **Infectious agents (NCBI Taxonomy):** *Cryptosporidium parvum* (NCBI:txid5807) / *C. hominis* (NCBI:txid237895) — biliary/intestinal; **Cytomegalovirus** (HHV-5, NCBI:txid10359); plus recurrent pyogenic bacteria and fungi (incl. *Pneumocystis*). These are **opportunistic consequences** of the immunodeficiency, not primary causes. **[human clinical]**

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6. Mechanism / Pathophysiology

Ordered causal chain

1. **Biallelic *IL21R* loss-of-function mutation** (e.g., p.Arg201Leu) → **leads to** absent or mistrafficked IL-21R protein that fails to reach/function at the plasma membrane and cannot bind IL-21. **[in vitro, demonstrated]**

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2. Loss of surface IL-21R → **prevents** assembly of the IL-21R/γc receptor complex and **abrogates** JAK1/JAK3 activation → **results in** failed phosphorylation of STAT3 (and STAT1/STAT5). **[in vitro, demonstrated]**

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 pathway review
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3. Absent STAT3 signaling → **fails to up-regulate the plasma-cell master regulator BLIMP1/PRDM1 (and to down-regulate BCL6)** → **impairs** IL-21-driven B-cell activation, immunoglobulin class-switch recombination, and differentiation into plasmablasts/plasma cells → **causes** hypogammaglobulinemia, poor specific-antibody responses, and reduced memory B cells. **[in vitro + human clinical]**

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— the latter: "*stimulation with IL-21 ... induced robust and prolonged STAT3 activation in primary human B cells*" and STAT3 "*triggered BLIMP1 mRNA and protein up-regulation, plasma cell phenotypic features, and Ig secretion*")

4. **Branch A (humoral/atopy):** Loss of IL-21's normal restraint on IgE, unopposed IL-4/Th2 activity → **leads to** elevated IgE, asthma, eczema, anaphylaxis. **[model organism + human clinical]**

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 high IgE/low IgG1 in Il21r^{-/-} mice;
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 IgE elevated in 50%)

5. **Branch B (cellular cytotoxicity):** Absent STAT3 downstream of IL-21R → **reduces** CD8⁺ T-cell memory and lytic-machinery induction, NK-cell cytotoxicity, and MAIT/NKT-cell numbers → **impairs** control of intracellular/opportunistic pathogens. **[human clinical/in vitro]**

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6. Combined humoral + cytotoxic failure + reduced Tfh help → **permits** chronic **Cryptosporidium** infection of intestinal and **biliary epithelium**. [human clinical]

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7. Persistent biliary *Cryptosporidium* → **drives** chronic inflammation of bile ducts → **causes** sclerosing cholangitis, fibrosis, cirrhosis, portal hypertension, end-stage liver disease (and rare cholangiocarcinoma risk by analogy to other cryptosporidial-cholangitis PIDs). [human clinical]

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8. Chronic immune dysregulation/impaired tumor surveillance → **contributes to** lymphoproliferation/lymphoma (e.g., marginal zone lymphoma). [human clinical, inferred]

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Detail by category

- **Molecular pathways:** IL-21 → IL-21R/γc → **JAK1/JAK3** → **STAT3** (canonical), with STAT1/STAT5 and PI3K-AKT and MAPK as secondary arms (Reactome "Interleukin-21 signaling"; KEGG "Jak-STAT signaling pathway hsa04630"). **GO:0038114** IL-21-mediated signaling pathway; **GO:0007259** JAK-STAT signaling.
- **Cellular processes:** B-cell differentiation (**GO:0030183**), isotype/class switching (**GO:0045190**), plasma-cell differentiation (**GO:0002317**), germinal-center/Tfh help, NK-mediated cytotoxicity (**GO:0042267**), CD8 memory formation.
- **Protein dysfunction:** Receptor **misfolding/mistrafficking and loss of ligand binding** (not aggregation); pure loss of function.
- **Immune system involvement:** Combined immunodeficiency + immune dysregulation (atopy, lymphoproliferation).
- **Tissue damage:** Immune-mediated + infection-driven **biliary fibrosis/sclerosis** and hepatocellular injury.

- **Cell types (CL):** B cells CL:0000236 (memory B, plasmablast CL:0000980), Tfh CL:0002038, CD8 memory T CL:0000909, NK CL:0000623, MAIT CL:0000940, NKT cells; biliary epithelial (cholangiocyte) as the injured target.
- **Omics:** No dedicated transcriptomic/proteomic/metabolomic disease signatures published (ultra-rare). Mechanistic data are from targeted immunophenotyping and STAT-phosphorylation flow assays. **[human clinical/in vitro]**

7. Anatomical Structures Affected

- **Organ level (primary):** **Bile ducts** (UBERON:0002394; intra- and extrahepatic — sclerosing cholangitis, bilateral/diffuse), **liver** (UBERON:0002107), **intestine/GI tract** (UBERON:0000160), **immune/lymphoid organs** (bone marrow UBERON:0002371, lymph node UBERON:0000029, spleen UBERON:0002106, thymus). **[human clinical]**
- **Secondary/complications:** lung/respiratory tract (UBERON:0002048; recurrent pneumonia incl. CMV/*Pneumocystis*), skin (UBERON:0002097; eczema), portal hypertension complications.
- **Body systems:** immune/hematolymphoid, hepatobiliary/digestive, respiratory, integumentary.
- **Tissue/cell level:** biliary/intestinal **epithelium** (infection target); lymphoid cell populations (B/T/NK as above).
- **Subcellular (GO CC):** plasma membrane (**GO:0005886**) — site of the receptor-localization defect; secretory pathway/ER involved in receptor mistrafficking.
- **Localization/lateralization:** Cholangitis is typically **diffuse/bilateral intrahepatic ± extrahepatic**; systemic immune involvement.

8. Temporal Development

- **Onset:** Pediatric/early childhood; **median 2.5 y (0.5–7 y)**; onset often **insidious** (chronic diarrhea, recurrent infections)

(P 33929673).

[human clinical]

- **Progression:** Hepatobiliary disease is **chronic and progressive** (cholangitis → fibrosis → cirrhosis → end-stage liver disease). Infections are recurrent; atopy episodic. Rate variable between patients.
- **Course/duration:** Chronic, lifelong without curative HSCT; frequently fatal in childhood if untreated or if transplant occurs after advanced organ damage.
- **Critical period:** A key **therapeutic window** exists — HSCT before advanced cholangiopathy/liver damage markedly improves outcome

(P 30850087).

[human clinical]

9. Inheritance and Population

- **Epidemiology:** Ultra-rare; ~20+ patients reported worldwide by 2021; no reliable prevalence/incidence figures. [human clinical]

(P 33929673)

- **Inheritance:** Autosomal recessive; biallelic *IL21R* mutations. [human clinical]

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- **Penetrance/expressivity:** Immunodeficiency penetrance appears **complete** in biallelic-null individuals; **expressivity variable** (onset age, presence of cholangitis/atopy/lymphoma).
- **Anticipation / germline mosaicism:** Not applicable/none reported.
- **Founder effects:** None established (8 distinct mutations/8 families).
- **Consanguinity:** Strong association — most families consanguineous.
- **Carrier frequency:** Not defined; alleles are private/ultra-rare in gnomAD → cascade (family-specific) rather than population carrier screening.
- **Demographics:** Reported across multiple ethnic groups (Middle Eastern, Turkish, European and others via 7 centers); **no sex predilection** (autosomal); age distribution skewed pediatric.

10. Diagnostics

- **Genetic testing (definitive): Whole-exome sequencing or NGS inborn-errors-of-immunity gene panels** identify biallelic *IL21R* variants; targeted single-gene/Sanger for cascade testing. WGS useful for non-coding/structural variants. CMA/karyotype/FISH not indicated. **[human clinical]**

(P 33929673):

5 novel patients found by "exome or NGS panel sequencing")

- **Functional confirmation (in vitro):** absent surface IL-21R expression; **abrogated IL-21-induced STAT3 phosphorylation** by flow cytometry; impaired IL-21-driven B-cell class switch/Ig secretion

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[in vitro]

- **Laboratory (LOINC-type):** serum immunoglobulins (hypogammaglobulinemia; **high IgE**), specific antibody titers, lymphocyte subsets (low memory B, low Tfh/MAIT, altered NK), liver function/cholestatic panel.
- **Microbiology: Cryptosporidium stool/bile PCR (more sensitive than microscopy)** — critical and often missed

(P 12690272):

"*Cryptosporidium* could be detected by PCR but not by microscopy").

- **Imaging: MRI/MRCP** or cholangiography for sclerosing cholangitis (ductal strictures/beading, dilatation); ultrasound.
- **Biopsy/pathology:** liver biopsy showing sclerosing cholangiopathy/fibrosis.
- **Clinical criteria / differential:** No disease-specific criteria; diagnosed under IUIS CID framework. **Differential:** IL-21 (ligand) deficiency (phenocopy;

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CD40L/CD40 (hyper-IgM), MHC class II deficiency, DOCK8 deficiency, STAT3-LOF hyper-IgE syndrome, CVID — distinguished by genetics + the IL-21-signaling assay.

- **Screening:** Not on newborn-screening TREC panels (patients usually have T cells). Carrier/cascade testing and prenatal/PGD available for known familial variants. **[human clinical]**

11. Outcome / Prognosis

- **Survival/mortality:** Poor without early curative therapy. In the largest cohort, **post-HSCT overall survival was only 33.3% (2/6)**, and mortality among non-transplanted patients was high

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In a broader PID sclerosing-cholangitis cohort, **7/13 (53.8%) died a median 4 months post-HSCT**, while 6/13 with milder cholangiopathy survived and improved

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[human clinical]

- **Prognostic factors:** Pre-existing organ (liver/biliary) damage is the key negative prognostic factor

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"pre-existing organ damage constituting a negative prognostic factor"); earlier diagnosis/transplant improves outcome.

- **Morbidity/QoL:** growth failure, chronic liver disease/portal hypertension, transplant-related complications; substantial disability.
- **Complications:** end-stage liver disease, portal hypertension, disseminated/opportunistic infection, lymphoma/lymphoproliferation, post-transplant *Cryptosporidium* recrudescence.
- **Recovery potential:** Immune reconstitution and cholangiopathy reversal possible if HSCT precedes advanced fibrosis.
- **Prognostic biomarkers:** severity/stage of cholangiopathy on imaging/histology; persistent *Cryptosporidium* positivity.

12. Treatment

- **Definitive (curative):** Allogeneic hematopoietic stem cell transplantation (HSCT) — NCIT:C15431; ideally with **reduced-intensity conditioning**, performed **before** advanced liver disease

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recommends *"early diagnosis and allogeneic hematopoietic stem cell transplantation"*;

(P 30850087).

[human clinical]

- **Sequential liver + HSCT (NCIT:C15393 liver transplantation):** for end-stage liver disease with combined immunodeficiency; curative in analogous PID cryptosporidial cirrhosis

(P 29377874).

[human clinical]

- **Supportive pharmacotherapy:**
- **Immunoglobulin replacement therapy** (IVIG/SCIG) — NCIT:C603 — for hypogammaglobulinemia.
- **Anti-*Cryptosporidium* agents:** **nitazoxanide** (CHEBI:189102/NCIT), **paromomycin**; often only partially effective without immune reconstitution.
- **Antimicrobial prophylaxis:** e.g., trimethoprim-sulfamethoxazole for *Pneumocystis*; antivirals/antifungals as indicated.
- **Atopy management:** asthma controllers, anaphylaxis precautions.
- **Advanced/experimental:** No approved gene therapy or targeted molecular therapy; *IL21R* is a plausible future **gene-addition/gene-editing** target by analogy to other γ c-family SCID gene therapies

(P 20660403).

No IL21R-specific clinical trials (NCT) identified. [human clinical/inferred]

- **Pharmacogenomics:** none specific.
- **Strategy:** Diagnose early → prevent/treat *Cryptosporidium* + IgG replacement + prophylaxis → **proceed to HSCT before liver damage**; combined liver+HSCT if cirrhotic.

13. Prevention

- **Primary prevention:** Not preventable at the genetic level; **genetic counseling** for consanguineous/at-risk families; **preimplantation/prenatal genetic diagnosis** for known familial variants.

- **Secondary prevention:** Early molecular diagnosis (NGS) in infants with recurrent infection/chronic diarrhea; **surveillance for *Cryptosporidium*** (PCR) and hepatobiliary monitoring (MRCP/LFTs).
- **Tertiary prevention:** IgG replacement, antimicrobial prophylaxis, timely HSCT to prevent progression; post-transplant vigilance against *Cryptosporidium* recrudescence.
- **Environmental/public-health: Water safety/filtration and hygiene** to avoid *Cryptosporidium* exposure (boil/filter water, avoid recreational water risk).
- **Immunization:** standard non-live vaccines as able; **avoid live vaccines** in combined immunodeficiency.
- **Counseling:** autosomal-recessive 25% recurrence risk per pregnancy; cascade carrier testing. **[human clinical/guideline-inferred]**

14. Other Species / Natural Disease

- **Taxonomy/orthologs:** *IL21R* is conserved in mammals. **Mouse** *Il21r* (NCBI Gene 60504; chr 7; NCBI:txid10090); rat, zebrafish orthologs exist.
- **Natural disease:** No well-characterized naturally occurring IL21R-deficiency disease in companion animals/wildlife (OMIA — none prominent). Veterinary relevance is chiefly as engineered research models.
- **Comparative biology:** IL-21/IL-21R humoral-regulatory function is evolutionarily conserved; the γ c-cytokine receptor family (IL-2/4/7/9/15/21) shares CD132 across species (structural conservation,

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[computational/model organism]

- **Transmission/zoonosis:** Not applicable to the genetic disease (though *Cryptosporidium* itself is zoonotic).

15. Model Organisms

- **Mouse (mammalian) — *IL21r*-knockout** (Ozaki et al. 2002, *Science*,

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normal lymphoid development but, after immunization, **higher IgE and lower IgG1; *IL4/IL21r* double-KO → dysgammaglobulinemia with severely impaired IgG. [model organism]**

- *Phenotype recapitulation*: reproduces human antibody dysregulation (IgE-high/IgG-low, impaired specific antibody) and demonstrates IL-21/IL-4 cooperation.
- *Limitations*: mice have grossly normal lymphoid development and **do not spontaneously develop cryptosporidial sclerosing cholangitis** (redundancy + differing pathogen exposure), so the hepatobiliary phenotype is not captured.
- *Applications*: dissecting IL-21 control of B-cell class switching, Tfh/germinal-center biology, CD8/NK function.
- **Other models**: γ c-family biology also modeled in zebrafish/other systems (e.g., IL-2R γ c SCID models,

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informing the shared-receptor mechanism though not IL21R-specific.

- **Genetic model types available**: knockout (constitutive); conditional/humanized IL21R models feasible but not disease-defining. Resources: MGI (mouse), Alliance of Genome Resources.

Supported vs. Refuted Hypotheses

Supported (evidence-backed): - Biallelic *IL21R* LOF causes an autosomal-recessive combined immunodeficiency

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- Mechanism is failed IL-21R→JAK/STAT3 signaling affecting B, Tfh, CD8-memory, NK, MAIT/NKT compartments

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- *Cryptosporidium*-driven sclerosing cholangitis/liver disease is the prognosis-defining complication

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- HSCT is curative but timing-dependent; organ damage worsens outcome

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- Il21r^{-/-} mice model the antibody dysregulation

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Refuted / not applicable: - Not X-linked (distinct from γ c/IL2RG SCID); heterozygotes unaffected → not dominant/dominant-negative. - No environmental/toxin primary etiology; no founder mutation; no somatic/oncogenic driver role.

Limitations & Future Directions

- Evidence rests on <25 **patients**; frequencies/prognosis have wide uncertainty. No prevalence, QoL (EQ-5D/SF-36), or omics datasets exist.
- MONDO/Orphanet identifiers should be verified against current releases.
- Future needs: international registry/natural-history study; standardized *Cryptosporidium* surveillance; evaluation of gene-editing/gene-addition therapy; optimized reduced-intensity conditioning and combined liver+HSCT protocols.

Key References (PMIDs)

23440042 (Kotlarz 2013, original description) · 33929673 (Cagdas 2021, cohort n=13) · 33966600 (Edeer Karaca 2021, marginal zone lymphoma) · 24746753 (Salzer 2014, IL-21 ligand deficiency) · 12446913 (Ozaki 2002, Il21r^{-/-} mouse) · 24126614 (Desjardins 2013, IL-21/B-cell memory review) · 23830147 (Ives 2013, CD8 memory) · 25941256 (Wilson 2015, MAIT/NKT via STAT3) · 30850087 (Hadžić 2019, HSCT for PID cholangitis) · 29377874 (sequential liver+HSCT) · 12690272 (Cryptosporidium PCR diagnosis) · 18354204 (Diehl 2008, IL-21/STAT3→BLIMP1 plasma-cell differentiation) · 37535730 (γc receptor structure) · 20660403 (γc-family gene therapy).

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